

B5.3 Feeding Buffers When Chains Are Equally Long

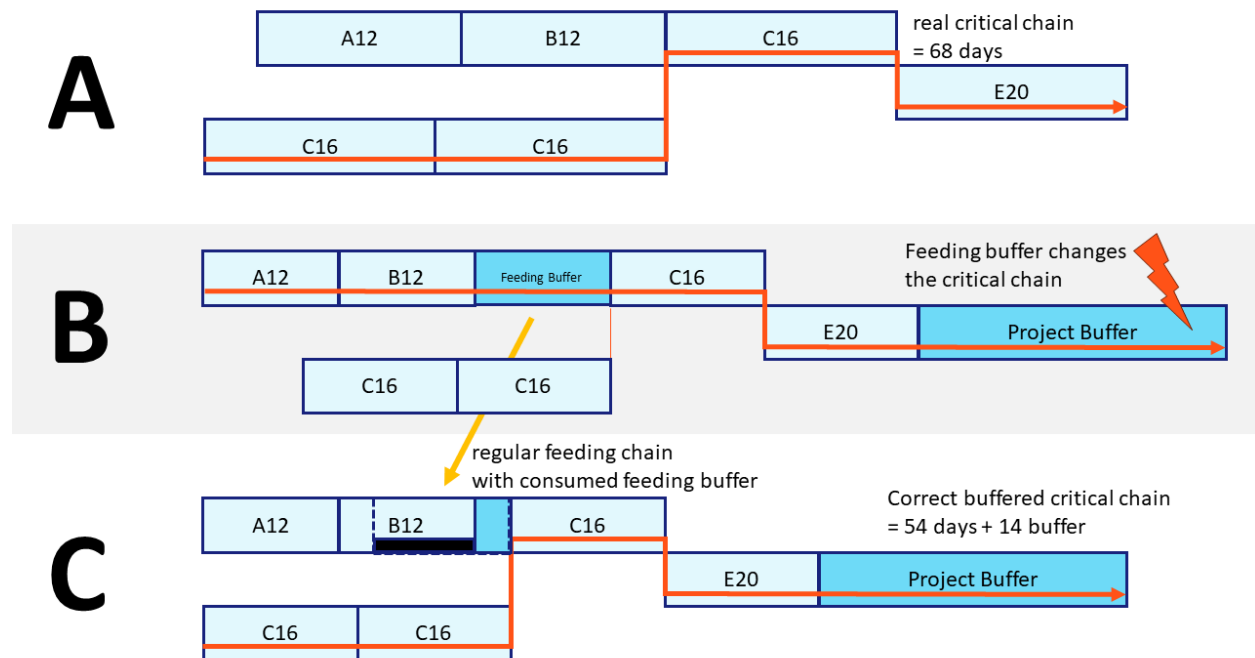


Figure 24. How to deal with feeding buffers when feeding chains are equally long? (A) shows a chain A12, B12 feeding into the C16, E20. (B) shows what happens if you add a 50% buffer as a feeding buffer - the critical chain elongates. (C) shows the right approach - the feeding buffer is preconsumed and the critical chain is untouched

In the realm of critical chain multi-project management, one of the key strategies employed is the use of feeding buffers. These buffers serve to protect the critical chain from potential delays caused by the feeding chains. To achieve this, the feeding chains are started a little early, and a block of time, known as the feeding buffer, is inserted to decouple the critical chain from each feeding chain.

The concept of feeding buffers is crucial in synchronizing the various paths to the critical chain. By subordinating everything else to the constraint, we ensure that the non-critical chains do not impede the progress of the critical chain. Feeding buffers act as a safeguard against unforeseen difficulties and disruptions that may arise in the non-critical chain tasks feeding into the critical chain. Additionally, they allow critical chain tasks to start early if circumstances permit.

Feeding buffers are strategically placed at integration points where the feeding chains meet the critical chain. This integration risk is reduced by securing the feeding chains, and daily status updates are provided in the form of easily understandable fever curves for all buffers. This enables the management team to have a clear overview of buffer consumption and progress, allowing them to take concrete measures to regain buffer if necessary.

The project buffer, which is another type of buffer in critical chain project management, is also established. However, it is important to note that the feeding buffer is separate from the project

buffer. The project buffer is specifically designed to protect the critical chain itself and requires the highest level of priority from project managers.

To further clarify the concept of feeding buffers, let's consider a project with one critical chain and several non-critical chains. The safety built at the task level is pooled together at the end of both the critical chain and non-critical chains. This aggregation principle allows for the averaging out of variations and risks, similar to how insurance aggregates multiple risks.

By reducing the total duration of the path, the feeding buffers contribute to a more efficient project timeline. They ensure that the critical chain remains the constraint and enable project managers to exploit its potential fully. Monitoring progress on the critical chain relative to buffer consumption becomes the primary focus of project management and controlling.

It is essential to actively manage deviations by strategically placing buffers at appropriate locations, such as before the integration point, at the end of feeding chains, and at the project's end. These buffers do not extend the project's duration but rather reallocate time from work package estimates. Critical Chain Project Management has proven to be a powerful approach for dealing with project management deviations in an agile manner.

While critical chain management is often associated with single projects, its true strength lies in portfolio or multi-project management. From a portfolio perspective, project management resembles production, with the goal of delivering as many projects as possible on time and with good quality.

In conclusion, feeding buffers play a crucial role in critical chain multi-project management. They protect the critical chain from potential delays caused by the feeding chains and allow for efficient synchronization of project paths. By actively managing deviations and strategically placing buffers, project managers can ensure the smooth progress of the critical chain and optimize project timelines.